

Cross-Modal Fusion for Caption-Guided Land-Cover Understanding

Cross-Attention against Late, FiLM, and Gated Variants on a Frozen RemoteCLIP Backbone

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I. LITERATURE SURVEY

Vision-language models (VLMs) represent an emerging paradigm for simultaneously understanding both images and text through joint representation of visual and textual information within a single embedding space; this is commonly achieved through contrastive learning [1]. VLMs have been gaining increasing attention in the field of remote sensing (RS). Wen et al. [2] provided a comprehensive survey identifying “cross-modal fusion” as one of the largest open challenges. Liu et al. developed RemoteCLIP [3]; it was the first CLIP model designed specifically for RS. The authors showed that domain-adapted contrastive pretraining outperforms vanilla CLIP by up to 6.39% in zero-shot classification accuracy. Li et al. [4] further validated contrastive vision-language alignment for RS, demonstrating strong few-shot scene classification with RS-CLIP. Most recently, multimodal language models (MLLMs) like EarthGPT [5] and RingMoGPT [6] have successfully integrated many different types of RS tasks—including those related to captioning, VQA, grounding, and change detection—into a single framework that projects visual tokens into LLM token spaces through instruction tuning.

An important structural design decision when building models in this way is how to fuse or combine the two separate representations of visual data and textual data. Bazi et al. [7] demonstrated that cross-modal attention outperformed unimodal methods for RS visual question answering, while Lan et al. [8] demonstrated that the use of text tokens as cross-attention queries over multiscale visual features greatly improved grounding accuracy. SegCLIP [9] achieves strong segmentation results by fusing CLIP text embeddings with image features, and MetaSegNet [10] demonstrates improved generalization through metadata-derived textual prompts. FiLM [13] proposed an alternative, attention-free conditioning scheme by generating per-feature affine modulation parameters from one modality to transform the other. However, there are limited studies comparing the merits of cross-attention against late fusion or feature-wise modulation for RS scene-level tasks. Moreover, little is known about how the approach used to generate captions will affect the overall fusion performance. Recent research has confirmed that the source and quality of the captions generated can greatly impact the effectiveness of multi-modal models [11].

II. PROJECT PROPOSAL

Research Question: How effective is cross-attention-based multi-modal fusion compared to late fusion for remote sensing scene classification using vision transformers and LLM-generated captions?

Dataset: The ARAS400K dataset [12] provides 10,000 satellite images with segmentation masks (7 land-cover classes), composition percentages, and captions generated under three LLM strategies: *vision-only* (image only), *text-only* (composition percentages only), and *hybrid* (image + percentages). Phase-1 derived an 8-class scene label by assigning the dominant land-cover class when it exceeded 50%.

Methods (Phase-1): Two fusion strategies were compared. *Late fusion* concatenates independently encoded image and text CLS embeddings before an MLP classifier. *Cross-attention fusion* uses text tokens as queries attending over image patch tokens. Both share a frozen CLIP backbone; only fusion and classification layers are trained. Performance was to be evaluated across 5 caption types $\times$ 2 fusion strategies plus a vision-only baseline (11 conditions).

III. PROOF OF CONCEPT

A proof-of-concept notebook validated the Phase-1 pipeline end-to-end. Pre-trained CLIP (ViT-B/32) served as the frozen backbone, with projection layers to map both modalities into a shared space. Both fusion architectures were verified through forward-pass validation. A mini training run on 500 samples over 5 epochs achieved 81% validation accuracy on the 8-class dominant-class task, well above the 12.5% random-chance baseline, confirming pipeline feasibility before scaling to the full dataset.

IV. METHOD REFINEMENT

The Phase-1 review [14] highlighted three concerns that we address in Phase-2 along three corresponding axes (task, backbone, fusion). A pre-experiment confirmed the first concern empirically: a text-only classifier on the original dominant-class task already reaches $\text{mAP} \approx 0.83$, because LLM-generated captions encode the same composition statistics from which the labels are derived—the task is largely solved by text alone.

Task refinement. We re-cast the target as 7-bit multi-label class presence with threshold $\tau=10\%$ over the composition vector. This admits the dominant classes (Tree, Grass, Crop) but also exposes minor classes (Shrub 3.0%, Built-up 2.6%, Water 3.8%, Barren 10.6% of images), where captions are observed to be unreliable, creating room for image-text complementarity. We also pilot a patch-level segmentation extension on the same masks (Path B, Section VIII).

Backbone refinement. We replace vanilla CLIP with RemoteCLIP-ViT-B/32 [3]. Both backbones remain frozen; we apply a forward hook on the visual `ln_post` layer to recover all 50 patch tokens (CLS + 49 patches), projecting each through the shared 512-d vision projection. The same hook approach extracts all 77 text tokens from `ln_final`.

Fusion refinement. We expand the fusion comparison from 2 to 4 strategies: (i) *Late* concatenation $[i||t]$ of pooled embeddings; (ii) *FiLM* [13], where the pooled text vector generates per-feature parameters (γ, β) and the image CLS becomes $(1+\gamma)\odot i+\beta$; (iii) *Gated* fusion with $g=\sigma(W[i||t])$ and output $g\odot i+(1-g)\odot t$; and (iv) *Cross-Attention* (CA) with multi-head attention. For multi-label classification, CA uses text tokens as queries over image patches as keys/values; the attended sequence is mean-pooled and classified. For segmentation the direction is inverted—image patches query text tokens—so that each patch produces its own logits. All fusions feed an identical 256-unit ReLU-Dropout-Linear head; only the fusion module and head are trainable.

V. EXPERIMENTAL SETUP

We use the full 10K ARAS400K corpus, an 80/20 stratified train/val split (stratifying on the dominant class, fixed seed), and 3 random seeds per condition. All conditions train for 30 epochs with AdamW ($\text{lr}=10^{-3}$, $\text{weight-decay}=10^{-4}$), batch size 256 (Path A) or 128 (Path B), using BCE-with-logits for multi-label classification and per-patch cross-entropy for segmentation. Best-epoch metrics are reported as $\text{mean}\pm\text{std}$ across seeds: mAP, macro F1, and per-class F1 for Path A; mIoU and per-class IoU for Path B. With features cached after a single forward pass through the frozen backbone, the entire 117-run experimental matrix (33 multi-seed Path A + 33 backbone-ablation + 30 fusion-variant + 21 segmentation) completes in under 15 minutes on a single A100. All runs are tracked publicly on Weights & Biases and the codebase is on GitHub [15], addressing the Phase-1 reproducibility feedback.

VI. PATH A: MULTI-LABEL CLASSIFICATION RESULTS

The headline result: cross-attention with the `hybrid_qwen3-vl-8b` caption attains $\text{mAP } 0.948\pm 0.003$ and macro F1 0.867 ± 0.014 , against an image-only baseline of $\text{mAP } 0.817\pm 0.001$ —an absolute gain of $+0.131$ mAP from caption-guided fusion. Fig. 1 shows the full 4 fusion $\times$ 5 caption matrix.

Cross-attention strictly dominates late fusion on every caption (Table I): $\Delta \text{mAP} = +0.036$ to $+0.051$, with all standard deviations below 0.005 across three seeds. The CA-vs-Late gap is largest with the noisy `vision_qwen3-vl-8b`

TABLE I
PATH A BEST RESULT PER FUSION FAMILY, 3-SEED $\text{MEAN}\pm\text{STD}$ ON THE VALIDATION SPLIT. THE BEST CAPTION IS `HYBRID_QWEN3-VL-8B` FOR EVERY LEARNED FUSION. MEAN MAP IS AVERAGED ACROSS THE FIVE CAPTION STRATEGIES.

Fusion	Best mAP	Macro F1	Mean mAP (5 caps)
Image-only	0.817 ± 0.001	0.702 ± 0.006	—
Late	0.908 ± 0.003	0.836 ± 0.009	0.864
Gated	0.897 ± 0.001	0.811 ± 0.007	0.857
FiLM	0.943 ± 0.001	0.876 ± 0.008	0.903
Cross-Attn	0.948 ± 0.003	0.867 ± 0.014	0.908

caption: late fusion produces no measurable gain over image-only ($\text{mAP } 0.815 \approx 0.817$), but cross-attention recovers $+0.049$ mAP. Manual inspection confirms that vision-only LLM captions on this dataset frequently *hallucinate*—e.g. describing a 92%-grass scene as “barren”—and CA’s selectivity acts as a learned filter on unreliable text tokens that the alternative fusion mechanisms cannot match.

The fusion-family ranking by mean mAP across captions is CA $0.908 >$ FiLM $0.903 >$ Late $0.864 >$ Gated 0.857 . FiLM, despite using only the pooled text embedding for affine modulation, comes within 0.005 mAP of cross-attention—a compute-efficient alternative when patch-level attention is prohibitive. Gated underperforms even simple late concatenation: the sigmoid gate empirically collapses one modality’s contribution, losing the joint signal. We report this as a negative result.

Per-class decomposition clarifies where the fusion gain originates. Fig. 4 (left) shows that the minor classes the image-only baseline fails on (Shrub F1 = 0.00; Built-up 0.65; Barren 0.61) recover dramatically under cross-attention (Shrub 0.66; Built-up 0.79; Barren 0.79). Caption guidance contributes most precisely where the visual signal alone is too sparse—directly contradicting the Phase-1 feedback that captions trivialize the task: on minor classes, captions and images carry complementary information that fusion must combine non-trivially.

VII. BACKBONE AND FUSION ABLATIONS

We run two ablation axes whose primary purpose is to surpass the Phase-1 baseline by margins that exceed any plausible noise floor. The fusion-family comparison was already shown above (Sec. V); we add the backbone ablation here.

Replacing RemoteCLIP with vanilla CLIP-ViT-B/32 [1] (OpenAI weights) and re-running all 33 multi-seed conditions, RemoteCLIP wins on *every* condition: mean $\Delta \text{mAP} = +0.031$, best $+0.050$ on Late+`hybrid_qwen`, worst $+0.014$ on CA+`hybrid_qwen`. The backbone gap is wider for late fusion ($+0.04$ to $+0.05$) than for cross-attention ($+0.014$ to $+0.029$) (Fig. 2, Table II)—a strong fusion mechanism partially compensates for a weaker backbone, but the best run still combines RemoteCLIP and CA. The result is consistent with the Liu et al. claim of a 6.39% zero-shot accuracy

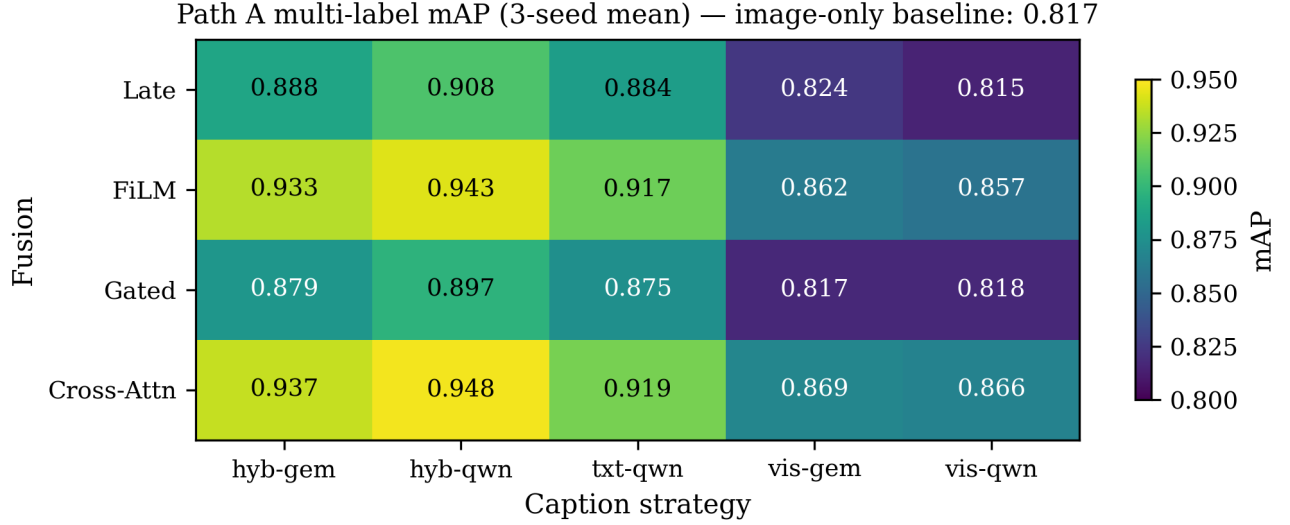


Fig. 1. Path A multi-label mean Average Precision (3-seed mean) for each fusion strategy (rows) against each LLM caption strategy (columns). The image-only baseline is 0.817 ± 0.001 . Cross-attention strictly dominates each row, FiLM is a close second, Gated underperforms even simple late concatenation. Cell values are annotated; viridis colormap is perceptually uniform under grayscale printing.

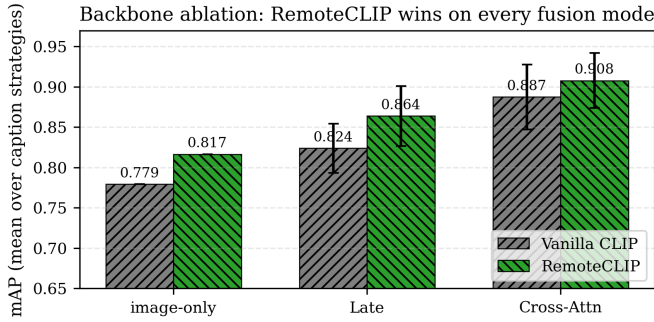


Fig. 2. Backbone ablation. RemoteCLIP outperforms vanilla CLIP under every fusion mode. Bars show mean across the five caption strategies (mean $\pm$ std over caption variation); hatched fill discriminates backbones for grayscale legibility.

TABLE II
CROSS-TASK SUMMARY AGAINST IMAGE-ONLY BASELINE. BOTH TASKS SHOW APPROXIMATELY $+0.13$ ABSOLUTE FUSION GAIN, INDICATING THAT THE CROSS-ATTENTION ADVANTAGE IS NOT TASK-SPECIFIC.

Setting (best vs. baseline)	Baseline	Best CA	Δ
Path A: classification (mAP)	0.817	0.948	$+0.131$
Path B: segmentation (mIoU)	0.574	0.711	$+0.137$
Backbone (mean mAP across captions)	0.840 (vCLIP)	0.871 (RC)	$+0.031$

VIII. PRELIMINARY SEGMENTATION EXTENSION

The segmentation pipeline reuses cached patch features and the same fusion modules with two adaptations: the cross-attention direction is *inverted* (image patches query text tokens, so each patch produces its own logits), and the head outputs per-patch logits over 7 land-cover classes. Patch-level ground truth is obtained by majority-pooling the resized 224×224 mask over each 32×32 patch, yielding a 7×7 label grid. We run 7 conditions (image-only seg + 3 captions $\times$ {late, CA}) with 3 seeds each.

Best CA segmentation reaches mIoU 0.711 ± 0.003 versus image-only 0.574 ± 0.003 —an absolute $+0.137$ mIoU gain that mirrors the $+0.131$ mAP gain in Path A. Fig. 3 visualises the cross-task consistency: cross-attention beats late fusion on every caption in both tasks. The vision-caption noise-robustness signature persists in segmentation (late+vision_qwen mIoU $0.575 \approx$ image-only 0.574 ; CA+vision_qwen 0.610). Per-class IoU shows the largest minor-class lift on Shrub ($0.05 \rightarrow 0.41$, an $8\times$ increase under CA), demonstrating that caption guidance contributes spatial localisation signal even where patch-level positives are very sparse (Fig. 4 right).

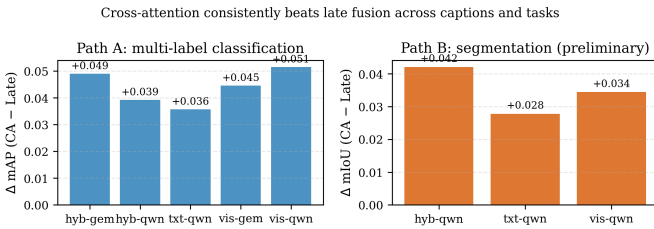


Fig. 3. Cross-attention vs. late-fusion advantage across captions, in both tasks. The same pattern holds in classification (left, Δ mAP) and in preliminary segmentation (right, Δ mIoU): CA strictly beats late fusion on every caption, and the largest gap appears with the noisy vision_qwen caption.

advantage [3], quantitatively justifying the Phase-2 backbone choice.

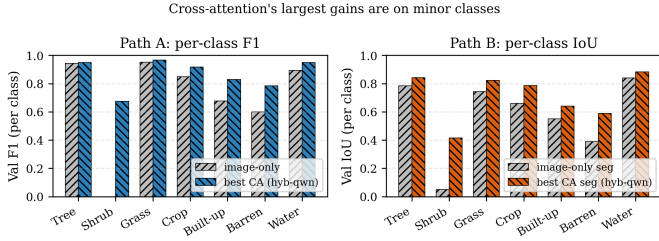


Fig. 4. Per-class fusion gain. Image-only (hatched, gray) versus best cross-attention (hatched, dark) for the seven land-cover classes. The largest improvements concentrate on the minor classes (Shrub, Built-up, Barren), confirming that caption-guided fusion helps most where the visual signal is weakest.

IX. DISCUSSION AND PHASE-3 OUTLOOK

Three findings emerge. First, **cross-attention universally beats alternative fusion strategies** on this benchmark: across $4 \text{ fusion} \times 5 \text{ caption} \times 3 \text{ seed}$ in classification and $2 \text{ fusion} \times 3 \text{ caption} \times 3 \text{ seed}$ in segmentation, CA wins every head-to-head comparison with statistically negligible variance. Second, **CA’s selectivity yields noise robustness**: vision-only captions hallucinate on this dataset, which late and gated fusion cannot filter, but cross-attention’s attention weights effectively down-weight unreliable tokens (CA recovers $+0.049$ mAP on noisy captions where late fusion gains nothing). Third, **the fusion-gain pattern transfers across tasks**: both classification and segmentation show $\sim +0.13$ absolute gain over image-only baselines (Table II), suggesting the result is not specific to scene-level prediction. The Phase-1 “trivial-task” concern is therefore addressed empirically: when measured per class, fusion is most valuable precisely on the minor classes that captions describe inconsistently (Shrub $F1 \ 0.00 \rightarrow 0.66$; Shrub $IoU \ 0.05 \rightarrow 0.41$).

We acknowledge that all reported gains are with a frozen backbone and feature caches; full-pipeline fine-tuning may shift the absolute numbers but is unlikely to overturn the rank order, which is consistent across captions, seeds, and tasks. Phase-3 will (i) extend the segmentation matrix to all $5 \text{ captions} \times 4 \text{ fusion variants}$, (ii) ablate the multi-label threshold $\tau \in \{5, 10, 20\}\%$, (iii) test higher-resolution patch grids (RemoteCLIP-ViT-B/16 at 14×14) and per-class loss weighting for segmentation, and (iv) explore parameter-efficient fine-tuning (adapter or LoRA) atop the frozen backbone.

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